POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Web Technology

Semester: Fall

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Discuss the need of web browser and web server in accessing the Internet. Also List the different MIME headers and explain them. 7

   b) Explain the essential difference between: 8

      i) URL and URI

      ii) HTTP and FTP

2. a) Why table-less design is preferred more than table-based design? Write HTML code for following : 8

Roll no:	Name	Subjects			
		C	Web	Math	C++
1	Abhisek	50	45	60	70
2	Suresh	50	45	45	NQ
3	Yodhin	80	85	90	75
Contact to department if Not Qualified(NQ)					

   b) Explain Standard HTML document Structure. How would you map image in HTML? 7

3. a) Create a HTML page with a div with some content and two paragraph tags with some content having ID P1 and P2. Write external css for the div tag having fixed position with border style solid and width 2px. The P1 should have relative position. The font type of P1 should be Arial and color should be green. The P2 have absolute position with top 0px and left 200px. 8

   b) How can you validate form in JavaScript? Write a program to explain it. Your program must contain email, password and submit button. On clicking, validate whether the fields are blank and the format for the email is right. 7

4. a) What do you mean by events and event handling? How will you locate mouse cursor using JavaScript? Explain with code. 2+5

   b) What is DOM? Write a code synopsis to represent dragging and dropping of elements in JavaScript. 2+6

5. a) How can you define variable in PHP? Define any two variable of string types and display their results after concatenation. 7

   b) How is array implemented in PHP? Differentiate between *foreach ()* loop and *each ()* function with example. 8

6. a) Make a login form with e-mail and password. Write a program to make dynamic login using php and MYSQL query. Your program must contain database connection file from php to MYSQL. 8

   b) Describe Starting, Storing and Destroying a Session. Explain with sample codes. Also, differentiate between Session and Cookies. 7

7. Write short notes on: (Any two) 2×5

   a) DOM Hierarchy in JavaScript

   b) Client side Vs server side scripting

   c) Ordered and unordered list

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Web Technology

Semester: Spring

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is hosting? Explain any four types of hosting. 8
   b) What is the purpose of MIME type specification in a request/response transaction between a browser and a server? 7

2. a) What are the different types of lists available in HTML? Explain each with suitable example code. 8
   b) Write HTML tags to generate the following form: 7

List with links	Image
Karnali ◦ Bardiya ◦ Banke ◦ Jumla	
Bagmati ◦ Kathmandu ◦ Bhaktapur ◦ Lalitpur	

3. a) What are the different levels of style sheets with their hierarchy of implementation? Explain the float and clear properties with a valid and detailed example. 7
   b) What are physical and logical tags? Give examples also. 4
   c) What do you mean by table-less design and how will you achieve it using CSS? 4

4. a) What do you mean by DOM in JavaScript? How do you get elements from document to JavaScript? Write an example of all methods. 7

b) Write a JavaScript program which validates the user data (name, email, contact no) and displays the success message if validated successfully. Also create the HTML form for entering the data. 8

5. a) What do you mean by events and event handling? Write HTML and JavaScript code to create and display a digital clock. 7

b) What are different file handling methods available in PHP? After validating the form values in Question 5 (a) write the contents to a 'users.txt' file with TIME of login. The content of file should look like: (no need to write the validation code in this answer). Also redirect user to home.php after writing contents to file. 8

Name: abc sharma, Email: testuser@gmail.com, Logged-in on: Fri Feb 01 2020 17:56:06 GMT+0545 (Nepal Time)

6. a) Explain the use of PHP $_POST , $_GET and $_REQUEST variables. 7

b) Write PHP-MySQL code validate the user log-in data (email and password). That is check whether the user data is available in database or not. Also in another file write code to display the user information (name and email). (Assume database name to be 'testdb' and table name 'users') 8

7. Write short notes on: (**Any two**) 2×5

a) Hyperlinks and hypermedia

b) Pop-up boxes in JavaScript

c) Types of array in PHP

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Web Technology

Semester: Fall

Year : 2022
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Let us assume a scenario, where person 'A' send an email to person 'B' and person 'B' response to person 'A'. Now, explain the sending and receiving process of email. 8

   b) What happens if you enter https://pu.edu.np into your browser? Describe the procedures behind the screens. Describe along with its block diagram. 7

2. a) Write HTML tags to generate the following design: 8

Labels	Input Fields
Full Name	
Gender	Male Female Others
Email	
Address	
Choose Color	
Upload Profile Picture	Choose file No file chosen
Register	

   b) Explain Standard HTML document Structure. Why table-less design is preferred more than table-based design? How grouping of elements is possible in HTML? 7

3. a) What are overriding rules in css? List out different types of css selectors. Explain box model in css with suitable figure. 7

   b) Write any 2 syntactic characteristics of JavaScript. Explain about different element access methods in JavaScript. 8

4. a) Why JavaScript is called dynamically typed Language? Explain with example how variables are declared in JavaScript. 5

b) Why JavaScript is called loosely typed Language? Explain with example how type conversion takes place in JavaScript. 5

c) Write a JavaScript program to input principal, time, and rate using prompt and display the calculated Simple Interest using alert 5

5. a) Write PHP inbuilt function to perform the following operations: 7

$list = [2, 6, 10, 8, 4 ,7]

$lang =["PHP" "JS", "C", "CPP"]

$days = [Mon"=>2, "Tue"=>3,"Wed"=>4"Thu"=>5];

i. Delete an individual element (8) from $list array.

ii. Assess last element from $lang array.

iii. Extract values from $days array.

iv. Extract keys from $days array.

v. Access third element from $list array.

vi. Sort $list in ascending order.

vii. Check if value (20) exists in $list array.

b) What are sessions and cookies? Mention any 5 differences between them with an example. An example must include all the functions of the session and cookies. 8

6. a) Develop a login form that includes Username and Password (Username= "admin" Password= "admin" already present in your database).Write a PHP and MYSQL code, first to verify log-in credentials. Second, when login is successful it navigates to the form that includes Name, Email, Phone, Password, and confirm Password. When you fill out the form it should display the information in the table structure. (Show database connection and assume database name and table name yourself) 8

b) Explain the use of PHP $_COOKIES, $_SESSION. How to write content in a particular file using PHP? 7

7. Write short notes on: (**Any two**) 2×5

a) Math Function in PHP (any 4)

b) Level of Style Sheets

c) DOM tree

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Web Technology | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

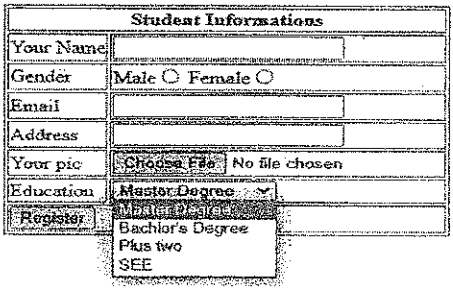
The figures in the margin indicate full marks.

Attempt all the questions.

1. a) If you are making a website for your company, what will you prefer, static website or dynamic website? Explain. 8

   b) What is a domain and fully qualified domain name? How can we register a domain name? Differentiate top-level and second-level domains. 7

2. a) Write HTML code to generate the following forms: 8



   b) Explain basic structure of HTML. Also illustrate display property and its different values. 7

3. a) Explain the different levels of style sheets with their hierarchy of implementation. What will be the actual height and width of content if padding, margin, border, content height, content width is 20px? Show using a box model. 8

   b) What do you mean by event and event handling ? Write a program that take number from the user and results its factorial after clicked on button "**CALCULATE**". 7

4. a) Add two new fields phone and password to Q2(a) and validates it with JavaScript. 8

Requirements:

i. All input fields should be filled else ***required** message should be displayed.

ii. Name should be of minimum length 3.

iii. Email should start with capital letter.

iv. Phone should be of length 10 and must be in format **XXX-XXX-XXXX[:where X = digit from 0-9]**

v. Password must start with a number and end with @.

b) Why are Constructors used in JavaScript? Describe the different types of element positioning and element visibility properties in JavaScript. 7

5. a) What are sessions and cookies? Mention any 5 differences between them with an example. An example must include all the functions of the session and cookies. Also, write a session program to count no. visits on the website. 8

b) Why PHP is called scripting language? How PHP actually works, along with HTML, CSS and JS? 7

6. a) Prepare a registration form to read student's informations (Name, Address, Age, e-mail and password) and save these informations in Database. Now make another page for login and display the informations after checking valid email id and password using PHP and MYSQL query. Make database name as "Pokhara" and table name as "info". 8

b) What do you mean by file handling? Write a PHP program to calculate area of rectangle and store the output into "data.txt". 7

7. Write short notes on: (**Any two**) 2×5

a) CSS Properties and Property values.

b) String function in PHP (any 5)

c) DOM model

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Web Technology

Semester: Fall

Year : 2023
Full Marks:100
Pass Marks: 45
Time :3hrs.

Candidates are required to give their answers in their own words as far as practicable.

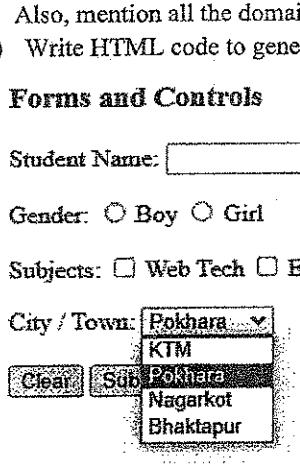
The figures in the margin indicate full marks.

Attempt all the questions.

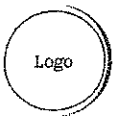
1. a) Define VPS(Virtual Private Server) hosting. How http protocol works. Explain with diagram. 7

   b) How to register a domain name? Explain root-level domains, top-level domains, second-level domains, and subdomains with their differences. Also, mention all the domains for https://www.pouexam.edu.np. 8

2. a) Write HTML code to generate the following forms: 7



   b) What are pseudo classes in CSS? Also design following logo using pseudo element. 8



3. a) What do you mean by DOM in JavaScript? How do you get elements 7

from documents? Write an example of all methods.

b) Write a JavaScript program which validates the user data (name, email, contact no) and displays the success message if validated successfully. Also create the HTML form for entering the data. 8

4. a) What do you mean by event and event handling? Write a program that take three numbers from the user using prompt and results greatest number after clicking on button "**I am Greatest**". 7

b) What are sessions and cookies? Mention any 5 differences between them with an example. An example must include all the functions of the session and cookies. 8

5. a) Why PHP called scripting language? How PHP actually works, along with HTML, CSS and JS? 7

b) What is the purpose of var_dump() and var_export() ? Define an Associative arrays and perform following operation sequentially : 8

i. Merge (arr1 and arr2)

ii. Sort(Key basis)

iii. Print all elements using foreach() loop

6. a) What are the applications of PHP? Explain the major features of PHP arrays. 7

b) Develop a login form that includes Username and Password (Username= "admin" Password= "admin" already present in your database). Write a PHP and MYSQL code, first to verify login credentials. Second, when login is successful it navigates to the file which asks for Your Name, Email, Phone, and Address. When you fill it out it should display the information in the table structure. (Show database connection and assume database name and table name yourself) 8

7. Write short notes on: **(Any two)** 2×5

a) Frameset VS iframe

b) String function in PHP(any 4)

c) DOM model

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Web Technology

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What do you mean by WWW? Write the difference between internet, intranet and extranet. 7

   b) Explain the difference between front-end and back-end web development. How does a web browser communicate with a web server to load a webpage? Also, mention key differences between a web application and a website. 8

2. a) Write the HTML code for the given design. 7

Roll no:	Name	Subjects			
		C	Web	Math	C++
1	Abhisek	50	45	60	70
2	Suresh	50	45	45	NQ
3	Yodhin	80	85	90	75
Contact to department if Not Qualified(NQ)					

   b) Explain the term HTML used in web-technology. What are the different types of tags used in HTML? How HTML differs from XHTML? 8

3. a) What is the function of selectors in CSS? Explain all sorts of selectors used in CSS with an example. 7

   b) Write a JavaScript program which validates the user data (name, email, contact no) and displays the success message if validated successfully. Also create the HTML form for entering the data. 8

4. a) What do you mean by DOM in JavaScript? How do you get elements from documents? Write an example of all methods. Also, mention the difference between querySelector and querySelectorAll. 7

   b) How events can be handled in JavaScript? Write a JS code for the following o/p. When the user provides a number from 1-7 it should return day. i.e(1-Sunday,2-Monday and so on) 5

Write Number of day of a week.

4	Try it

Today is Wednesday

c) Why JavaScript is called a loosely and dynamically typed language? Write one syntactic characteristics of JS. 3

5. a) Explain the use of PHP $_GET, $_POST, $_SERVER code in PHP with example. 7

b) $num=[2,6,10,8,4,7] $lang=["PHP", "JS","C", "CPP"] create the given array and 8

From the above array solve the following question

i. Check if value 20 exists in $num array.

ii. Access 3rd element from $num array.

iii. Delete an individual element 8 from the $num array.

iv. Combine $num and $lang array.

6. a) What are sessions and cookies? Mention any 4 differences between them with an example. Also, explain how $_SESSION can be used to manage the state on the web with login and logout. 7

b) Write PHP script to insert form data to database table on form submit. Where every data must be in lowercase before inserting into table. Also, develop the webpage as below: (No need to write CSS code) 8

Full Name

Email

Address

Submit

7. Write short notes on: **(Any two)** 2×5

a) Array methods in JS (Any 5)

b) MIME

c) Regular expression and pattern matching in PHP

d) Box model in CSS

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2024
Programme: BE | Full Marks: 100
Course: Web Technology (New) | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is the purpose of web servers in hosting websites and web browsers in rendering web pages according to the client-server model? Could you also explain the distinctions between front-end and back-end development, as well as the functional and content delivery techniques between static and dynamic websites? 7

   b) Explain the following terms: 8

      i. URL vs URI
      ii. MIME Headers
      iii. HTTP vs HTTPs
      iv. SMTP vs POP3

2. a) Explain types of CSS. Define background, position and border CSS properties. 7

   b) Write the html code for the following given design: 8

Fill the form below	
Name	
Password	
Feedback	
Gender	○ Male ○ Female
Subject	☐ Web ☐ Math ☐ Graphics ☐ English
	Reset All / Submit Above Details

3. a) Create HTML form that includes (Email, Phone, Password, and Confirm Password and submit button) and write a JS program that 7

validates the user data and displays the alert message if validated successfully.

OR

What is a regular expression? How does JS support Pattern matching? Provide some samples of Regx for Email, phone, date, visa card number, etc. Explain all the methods of Regx for pattern matching in JS with an example.

b) Synchronous and Asynchronous in JS. How Asynchronous is achieved in JS? 8

4. a) Why is Javascript called client-side scripting language? How are JS scripts included in the HTML page, explain with proper example. 7

b) Perform the following for a string= "Pokhara University" 8

i. Find the length of the string.
ii. Find whether the 'Uni' pattern is occurred or not.
iii. Display only Pokhara from the string.
iv. Find the Ascii value of character present at 4th index.

5. a) Why PHP called scripting language? How php actually works, along with HTML, CSS and JS? 7

b) What are sessions and cookies? Mention any 5 differences between them with an example. The example must include all the functions of the sessions and cookies. 8

6. a) What is the GET and POST method in PHP? How do they differ? Explain $_SERVER, $_POST, $_GET, $_REQUEST. 7

b) Develop a simple web page that asks for the user's login ID and Password if you provide the correct credentials, it will ask you to put your (name, email, phone, Nationality, and Marks) and store it in the database using a server-side script and also displays them in tabular form. Assume Database name: PU table name: Student. 8

OR

Write a PHP with MySQL code for the following. (Only MySQL code is not allowed. You have to show it with PHP as well)

i. Create a database with the name "Nepal"
ii. Connect with the database name "Nepal".
iii. Create a Table with name Family that includes (Name, Gender, Age, Qualification, DOB)
iv. Insert any 3 records.
v. Display those records in tabular form.

vi. Update any 2 records yourself
vii. Delete any 2 records yourself
viii. Drop Your database

7. Write short notes on: **(Any two)** 2×5
   a) Math function in PHP
   b) Rest and Spread operators
   c) Hyperlinks and Hypermedia
   d) Responsive Web Design

(WT-New - 2024 spring)

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Web Technology

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate static and dynamic websites with their applications and limitations of each. 8

   b) Explain the Domain Name System (DNS) and its hierarchical structure. How is a domain name registered, and what role does web hosting play in ensuring a website's accessibility on the Internet? 7

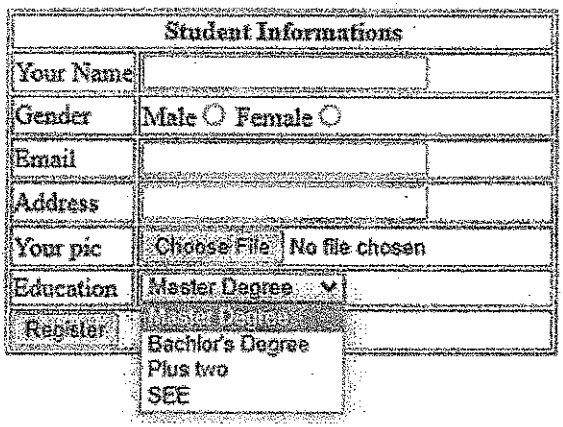
2. a) Explain the following aspects of HTML with examples: 8

   i. Images: The purpose of the <img> tag and its key attributes.

   ii. Multimedia: Methods for embedding audio and video in an HTML page.

   iii. Text Formatting: The use of <b>, <i>, <u>, <mark>, and <blockquote> tags for text formatting.

   b) Write HTML code to generate the following forms: 7



3. a) Why CSS is used in Web designing? Explain different levels (types) of style sheets with example. 8

	b)	How can you manipulate HTML elements using JavaScript? Provide examples to demonstrate adding, removing, and modifying content and attributes in the DOM.	7
4.	a)	Create a simple counter application. The page should display a number that starts at 0 and two buttons: Increase and Decrease. Clicking the buttons should update the number accordingly.	8
	b)	What is dynamic stacking? Write a program in JavaScript to show dynamic stacking of three images.	7
5.	a)	Write a program in PHP to list out all prime numbers within a given range.	8
	b)	Explain the origin, major features and arrays of PHP.	7
6.	a)	What are cookie and session in PHP? Explain the use of them with necessary program code.	8
	b)	How do you connect PHP with MySQL? Write a PHP code that retrieves the name and password from database.	7
7.		Write short notes on: **(Any two)**	2×5
	a)	Php file handling	
	b)	Compare var, let, and const	
	c)	Multimedia in HTML	

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Web Technology (New)

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

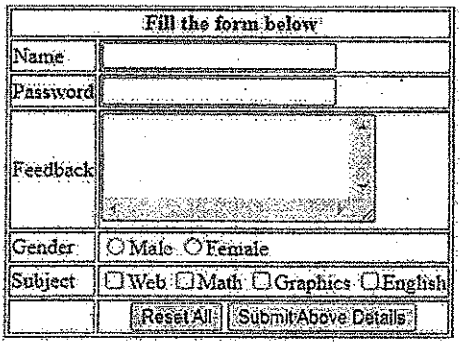
Attempt all the questions.

1. a) What is the purpose of web servers in hosting websites and web browsers in rendering web pages according to the client-server model? Could you also explain the distinctions between front-end and back-end development, as well as the functional and content delivery techniques between static and dynamic websites? 7

   b) What are web protocols? Explain the benefits of HTTPS over HTTP protocol. 8

2. a) How image is inserted in html? Explain the use of Image Maps with necessary program code. 7

   b) Write the html code for the following given design: 8



3. a) What are the types of CSS? Explain with necessary program code. 7

   b) Explain Synchronous and Asynchronous communication in JS. How is Asynchronous communication achieved in JS? 8

4. a) Create HTML form that includes (Email, Phone, Password, and Confirm Password and submit button) and write a JS program that validates the user data and displays the alert message if validated successfully. 7

OR

What is a regular expression? How does JS support Pattern matching? Provide some samples of Regx for Email, phone, date, visa card number, etc. Explain all the methods of Regx for pattern matching in JS with an example.

b) Why is JavaScript called client-side scripting language? Explain event handling in JavaScript with necessary program. 8

5. a) Why PHP called scripting language? How php actually works, along with HTML, CSS and JS? 7

b) What is the GET and POST method in PHP? How do they differ? Explain $_SERVER, $_POST, $_GET, $_REQUEST. 8

6. a) What are cookie and session in PHP? Explain the use of them with necessary program code 7

b) Develop a simple web page that asks for the user's login ID and Password if you provide the correct credentials, it will ask you to put your (name, email, phone, Nationality, and Marks) and store it in the database using a server-side script and also displays them in tabular form. Assume Database name: PU table name: Student. 8

OR

Write a PHP with MySQL code for the following. (Only MySQL code is not allowed. You have to show it with PHP as well)

i. Create a database with the name "Nepal"
ii. Connect with the database name "Nepal".
iii. Create a Table with name Family that includes (Name, Gender, Age, Qualification, DOB)
iv. Insert any 3 records.
v. Display those records in tabular form.
vi. Update any 2 records yourself
vii. Delete any 2 records yourself
viii. Drop Your database

7. Write short notes on: **(Any two)** 2×5

a) Session Handling

b) Rest and Spread operators

c) Php string

d) Responsive Web Design

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2021
Programme: BE Full Marks: 100
Course: Database Management System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Data Independency. Differentiate between Schema and Instance with the help of an example. 7

   b) Differentiate between Data model and E-R model. Draw an E-R diagram for a Library Management System including primary key, weak entity, composite attribute, derived attribute and multivalued attributes in your ER diagram. 8

2. a) Consider following relations, where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries: 8

   Doctor(SSN, FirstName, LastName, Specialty, YearsOfExperience, PhoneNum) Patient(SSN, FirstName, LastName, Address, DOB, PrimaryDoctor_SSN) Medicine(TradeName, UnitPrice, GenericFlag) Prescription(Id, Date, Doctor_SSN, Patient_SSN) Prescription_Medicine(Prescription Id, TradeName, NumOfUnits)

   i. List the trade name of generic medicine with unit price less than $50.
   ii. List the first and last name of patients whose primary doctor named 'John Smith'
   iii. List the first and last name of doctors who are not primary doctors to any patient.
   iv. List the SNN of distinct patients who have 'Aspirin' prescribed to them by doctor named 'John Smith'.

   b) Write SQL statements for following: 7

   i. Create a table named Chef with chef_license as primary key and following attributes:
   chef_license, c_fname, c_lname, c_dob, c_gender, c_experience_hours, c_photograph
   ii. Enter a full detailed information of a chef.
   iii. Change chef's experience hours by any value.

iv. Remove all chef records whose name contains character 'r' in second position in his first name.

v. Display the total experience hours of all chef.

vi. Create a view from above table.

vii. Display details of chef ordering on descending manner in last name and by first name when last name matches.

3. a) Explain Data Constraints and its types with examples. 8

b) What is the role of Triggers? Write an SQL trigger to carry out the following action: On **delete** of an account, for each owner of the account, check if the owner has any remaining amount, and if she does not, delete her from the *depositor* relation. 7

4. a) Explain about database normalization and its importance. Explain 1NF, 2NF and 3NF with examples. 8

b) What is security violation and integrity violations? Explain the need of access control, Authorization and Authentication. 7

5. a) Define query optimization. What are the basic steps of query processing with the help of a diagram 8

b) How do you evaluate the performance of a magnetic disk? What are the optimization techniques to reduce the disk block access? 7

6. a) What is a transaction? Write about the 'ACID' property of any transaction. 7

b) What is Crash Recovery? Explain log based recovery and what happens for a log-based recovery. 8

<T0 Start >

<T0, A, 1000,950 >

<T0, B, 2000, 2050 >

<T0 commit >

<T1 Start >

<T1, C, 700, 600 >

7. Write short notes on: **(Any two)** 2×5

a) RAID

b) Stored procedure

c) Distributed Database

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2021
Programme: BE | Full Marks: 100
Course: Database Management System | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is data independency? Why is it required in DBMS? Explain in detail. 7

   b) What is ER diagram? Draw an ER diagram for a library system. Assume the entities- student, teacher, book and semester. In the diagram illustrate the concept of strong entity, week entity, composite attributes, multivalued attributes, and derived attributes. 8

2. a) Define schema and views. Considering the following schemas: 8

   Sailors (sid, sname, rating, age)

   Boats (bid, bname, color)

   Reserves (sid, bid, day)

   Write relational algebra expressions for the following queries:

   i. Find the records of sailors who have reserved boat number 103 (bid=103).
   ii. Update the color of the boat, where bid is 104, into green.
   iii. Find the names of sailors who have reserved a red or green boat.
   iv. Find the names of sailors who have reserved boat number 103 on day 5.
   v. Find the names of sailors whose name is not 'Ram'
   vi. Find the names of all boats.

   b) What are DDL and DML queries in SQL? Consider the relations in 2(a) and write the SQL statements for the queries in 2(a). 7

3. a) What are the different types of integrity Constraints? Explain with examples. 8

b) What is database normalization? Discuss normalization process with a suitable example until it satisfies 3 NF. 7

4. a) What are the needs of security? Explain about the access control, authorization and authentication. 7

b) Consider the relation schema in 2(a). Write the relational algebra expression for the query "Find the names of sailors who have reserved a red or green boat". Construct the initial operator tree and final efficient operator tree after applying transformation rules. 8

5. a) Explain file organization using hash indices with example. 7

b) What is Crash Recovery? Explain log based recovery method with example. 8

6. a) Explain the serial schedule and serializable schedule with examples. 8

b) What are object- oriented database model? Explain the advantage and disadvantage of object-oriented database over relational database. 7

7. Write short notes on: **(Any two)** 2×5

a) Data dictionary

b) ACID properties

c) Query By Example (QBE)

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2022
Programme: BE Full Marks: 100
Course: Database Management Systems Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define DBMS. What are the advantages of DBMS over traditional file system? 7

   b) Why do you need E-R diagram? Draw an E-R diagram for online shop management system. Assume relevant entities and attributes for the given system. 8

2. a) Suppose we have the following relation: 8

   Employee(person_name, street, city)

   Works (person_name, company_name, salary)

   Company (company_name, city)

   Write relational algebra expressions for the following queries:

   i. Find names of all employees who live in 'Butwal' and whose salary is less than Rs. 50,000.

   ii. Find the names of all employees who work for "Nabil Bank Limited".

   iii. Find the names and cities of residence of all employees who work for "Global bank"

   iv. Update the salary of all employees by 10%.

   b) Define stored procedure. What is the advantage of the stored procedure? Explain how stored procedures are created in SQL. 7

3. a) Consider the relation **Actress_Details** and write the SQL statements for the following queries: 8

Players_id	Actress_name	Debut_year	Recent_release	Actress_fee
1	Renu	2010	Samay	400000
2	Sita	2022	Radha	300000
3	Geeta	2001	Mato	600000
4	Amita	1990	Man	700000
5	Karishma	1989	Prem	100000

i. Create the table Actress_details relation.
ii. Delete the data of actress whose recent release is Prem.
iii. Modify the database so that Renu's new release is "Win the Race" film
iv. Insert a new record in the above table.

b) Consider the following relation where {M_ID and P_ID} are primary keys. State in which normal form the relation is. What anomalies can occur in this relation? How can these anomalies be removed? 7

M_ID	M_Date	P_ID	Quantity
M11	16 June, 2022	I1	20
M11	26 June, 2022	I6	30
M22	3 September, 2022	I5	20
M22	13 September, 2022	I6	60
M22	23 September, 2022	I2	35

4. a) When do you use triggers? Explain with any one example of triggers in SQL. 7

b) Differentiate between authentication and authorization. How encryption and decryption occurs in private key and public key cryptography? 8

5. a) Consider the relation schema in 2(a), Write the relational algebra expression for the query "Find the names of all employees who lives in Pokhara". Construct the initial operator tree and final efficient operator tree after applying transformation rules. 8

b) What is file organization? Explain how you organize files using hash index. 7

6. a) What is crash recovery? Discuss shadow paging with necessary diagram. 7

b) What do you understand by concurrency control? Explain two phase locking protocol with examples. 8

7. Write short notes on: (**Any two**) 2×5

a) Remote backup system
b) ACID properties
c) Distributed database

POKHARA UNIVERSITY

Level: Bachelor | Semester : Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Database Management System | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between database schema and instances. Briefly describe DDL, DML and DCL. 8

   b) Define relational data model. Draw an E-R diagram for a Library Management System including primary key, weak entity, composite attribute, derived attribute and multivalued attributes in your ER diagram. 7

2. a) Suppose we have the following relation. 8

   Employee(person_name, street, city)

   Works (person_name, company_name, salary)

   Company (company_name, city)

   Write relational algebraic expressions for the following queries:

   i. List the name and city of employee who work in "pokhara" and have salary greater than Rs. 50,000.

   ii. Find the names of all employees who work for "ABC bank".

   iii. Delete all employee who come from "Chitwan".

   iv. Increase salary of all employee by 15%.

   b) What are different kinds of joins? Explain in brief. 7

3. a) Write SQL statements for the following queries using the given Employees relation: 8

E_id	Fname	Lname	Department	Salary	Hire_Date
01	Ramu	Bashyal	Sales	20000	2023-08-08
02	Damu	Pandey	IT	50000	2022-01-01
03	Biru	B.K.	Sales	40000	2021-02-10
04	Hiru	Dhamala	HR	35000	2023-12-18
05	Biren	Khadka	IT	60000	2012-10-22

i. Create a database named Company and Employees relation.

ii. Create a view that shows the E_id, Department and Hire_Date of all employees.

iii. Modify the table such that the Department of Biren is HR now.

iv. Delete the record of employees whose Lname is "Pandey".

b) What is referential integrity? Explain about a trigger with an example. 7

4. a) What is database normalization? Explain in detail about INF, 2NF, 3NF with suitable examples. 8

b) What are authorization and authentication? Why are they important? Explain in detail. 7

5. a) What are the steps in query processing? Make an operator tree for the following SQL expression: 7

```
Select customer_name
FROM branch, account, depositor
WHERE branch_city='btl' AND balance>2000;
```

b) What are the benefits of using B Tree index over the sequential and indexed sequential file organization? Explain. 8

6. a) Explain Log based recovery system with an appropriate log record example. 7

b) Why should the transactions' schedule be serialized? Explain conflict and view serializability with example. 8

7. Write short notes on: (Any two) 2×5

a) Data Dictionary

b) Stored procedure

c) Object oriented Database

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Database Management System (Old)

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why do you need DBMS? Explain in detail. 7

   b) Draw an ER diagram of a college database that keeps the information about student, teachers, course, department and semester. (Assume suitable relationships and attributes by yourself). 8

2. a) Consider the following relations: 8

   STUDENT(Student_ID, Student_Name, Major)
   COURSE(Course_ID, Course_Name, Credits)
   ENROLL(Student_ID,Course_ID,Grade)

   Write the relational algebra expression for:

   i. Find the names of all students who are enrolled in a course with a course name that starts with "MATH".
   ii. Update the credits of courses to 4 where the credits are currently less than 3.
   iii. Find the average grade for each major.
   iv. Update the name of the "COMPUTER_SCIENCE" major to "COMPUTER_ENGINEERING".

   b) What are triggers? Explain with suitable examples. 7

3. a) Consider the following relations: 8

   Employee(emp_id, name, address, telephone, salary, age)
   Works_on (emp_id, project_id, join_date)
   Project(project_id, project_name, city, duration, budget)

   Write the SQL commands for the following:

   i. Find the name of projects whose name starts with m and ends with a.
   ii. Find the project name with maximum budget.
   iii. Delete information of employees whose age is greater than average age of all employees.

iv. Increase salary of employee who lives in Butwal by 25%.

b) What are the problems in the given relation? Normalize this relation up to 3 NF to remove those problems. 7

SID	Name	SubID	SubName	FeePaid
1	Hari Dangol	ENG	English	20000
2	Mohan Shah	ENG	English	20000
3	Indira Rimal	ENG	English	30000
1	Hari Dangol	CPROG	C programming	20000
2	Mohan Shah	CPROG	C programming	15000
3	Indira Rimal	MATH	Mathematics	30000

4. a) What is a stored procedure? Explain with an example. 7

b) Consider the relational schema: 8

Employee(person_name,street,city)

Works(person_name,company_name,salary)

Company(company_name,city)

Write the relational algebra expression for the query "Find the names of all employees whose company is located in Pokhara". Construct an initial operator tree and final efficient operator tree after applying transformation rules.

5. a) Differentiate between authentication and authorization in brief. 7

b) What is file organization? Explain how you organize files using hash index. 8

6. a) Explain the log-based recovery in detail with suitable example. 7

b) What is serializable schedule? Explain conflict serializability and testing of conflict serializability with suitable examples. 8

7. Write short notes on: **(Any two)** 2×5

a) ACID properties

b) Distributed database

c) Referential Integrity

POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BE | Full Marks: 100
Course: Database Management System (New) | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is data abstraction in DBMS? Explain in detail. 7

   b) Consider you are asked to design a database for the Exam section of your college. Draw its ER diagram assuming required entities and their attributes. 8

2. a) Convert the ER diagram that you designed in question no 1 b) into relational schema. 7

   b) What are the views? Consider the table **tbl_emp** as follows: 8

Emp_id	Emp_name	Salary	Department	Date_of_joining
101	Anish	20000	Packing	2070-02-01
102	Rojina	18000	Cleaning	2075-04-06
103	Sita	35000	Polishing	2078-09-12

Write the SQL statements for the following:

i. Create the above table by considering Emp_id as primary key and insert the above records.
ii. Change the Department of Anish to marketing.
iii. Increase the salary of employees whose department is Cleaning by 12%.
iv. Find the name of employees having salary greater than 16000 and who joined after 2072-11-25
v. Add a new column Address to the above table.
vi. Delete the entire table.

OR

Why is joining in SQL necessary? Explain Inner Join, Natural Join and Outer Join with suitable examples.

3. a) What are the different types of integrity constraints? Explain with examples. 8

b) What is denormalization? Why is it necessary? Explain in detail. 7

4. a) What is the role of views in security? Explain the concept of authorization with suitable examples. 7

b) Explain the basic steps in query processing in detail. 8

5. a) Explain how the records are organized using fixed-length and variable-length records. 7

b) What is serializability and why is it needed? Explain the ACID properties in brief. 8

6. a) What is crash recovery? Explain log-based recovery method with example. 7

OR

What is transaction rollback? Explain how the Remote Backup System provides high availability and recovery facility.

b) List out the different categories of NoSQL databases. Explain the concept of blockchain with its properties. 8

7. Write short notes on: **(Any two)** 2×5

a) Nested Queries

b) Third Normal Form (3NF)

c) Lock-based protocols for concurrency control

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Database Management System

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define schemas and instances. What are the major functions of Database Manager? Discuss. 7

   b) Explain the benefits of data model. Construct an ER diagram for keeping records for Library Management System. 8

2. a) Consider the following schema: 8

   EMPLOYEE (Employee_ID, Employee_Name, Department)
   PROJECT(Project_ID,Project_Title, Budget)
   WORKS_ON(Employee_ID, Project_ID, Hours_Worked)

   Write the relational algebra expression to:

   i. Find the names of all employees who are working on a project with a project title that contains the word 'Development'.
   ii. Update the budget of projects to 1,000,000 where the budget is currently less than 500,000.
   iii. Find the total hours worked by each employee across all projects.
   iv. Update the name of the 'HR' department to 'Human_Resources'.

   b) Write the SQL statements for the following queries by reference of Hotal_details relation. 7

3. a) What are advantages of normalization? Illustrate importance of normalization with suitable examples. 8

   b) Why Integrity Constraints are used? Explain types of integrity constraints with examples. 7

4. a) Differentiate between authorization and authentication? Explain about private key cryptography with example. 8

   b) What is query processing? Explain its steps with block diagram. 7

5. a) Create a B+ tree of order 4 with the following data: (1,4,7,10,17,21,31,25,19,20,28) 7

b) Explain log based recovery and different methods of keeping log along with example. What is shadow paging? 8

6. a) What is view serialize ability? Explain 2PL and time stamping. 7

b) What are distributed database systems? Explain advantages and disadvantages of object oriented database model. 8

7. Write short notes on: (**Any two**) 2×5

a) Data Abstraction

b) ACID

c) Triggers

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Database Management System (New)

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is a database management system? Difference between Physical and Logical Data Independence. 7

   b) Draw E-R diagram for airline reservation system. The system must keep track of customers and their reservations, flights and their status, seat assignments on individual flights. Include appropriate relationship and cadinality. 8

OR

Draw an ER diagram of an "Exam System of Pokhara University" using extended features. Explain Primary key and Foreign key.

2. a) Consider the following schema: 7

   STUDENT(Student_ID, Student_name, Major)
   COURSE(Course_ID, Course_name, Credits)
   ENROLLS_IN(Student_ID, Course_ID, Grade)

   Write the relational algebra expressions for the following cases:

   i. Find the names of all students who are enrolled in courses with names that start with "C".
   ii. Update the grades of students to 'A' whose current grade is 'B'.
   iii. Find the average credits of courses each student is enrolled in.
   iv. Update the name of the course "Physics" to "Advanced Physics".

   b) Let us consider the following relation: 8

   Sailors (sid, sname, rating, age)
   Boats (bid, bname, color)
   Reserves (sid,bid, day)

   Write a SQL statements for the following:

   i. Find the records of sailors who have reserved boat number 75 (bid=75).

ii. Update the color of the yellow boat to green.

iii. Delete the records of sailors whose rating is less than 5.

iv. Find the name and rating of sailors who have reserved a black or blue boat.

v. Find the age of sailors who have reserved boat number 25 on day 3.

3. a) Define integrity constraints? Explain different integrity constraints with suitable examples. 8

OR

Why is database normalized? Explain about 1NF, 2NF, 3NF & BCNF.

b) Explain different types of functional dependencies with examples. Find the closure of attributes A, AC and ABC for the relation R(A,B,C,D) where A→B,BC→D are the dependencies in R. 7

4. a) Compare authentication and authorization. Explain the different types of access control mechanisms. 7

b) How query optimization is carried out? Explain about cost estimation of query. 8

5. a) Construct a B+ tree of order 4 for the given key values assuming empty tree initially and keys arranged in ascending order: {4,8,1,20,5,15,16,7,9,38,34,6,39,25} 7

b) What is serializability? Explain how concurrency control protocols are used? 8

6. a) What are NoSQL databases? Explain the advantages of NoSQL databases. 7

b) Differentiate between centralized and distributed databases. Explain in brief about NOSQL. 8

7. Write short notes on: **(Any two)** 2×5

a) Equivalence of expressions

b) Stored Procedures

c) ACID Property

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Database Management System

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Data Independency. Differentiate between Schema and Instance with the help of an example. 7

   b) Draw an E-R diagram for an online shopping system. Assume necessary entities and attributes. Also, the system should model the following: 8

   - A customer can place many orders, and each order contains many items.
   - An order can be shipped to different addresses, and each shipping address can be linked to multiple orders.
   - The system tracks payment status and the payment method for each order.

2. a) Using the following schema represent the following queries using Relational algebra: 8

   Doctors (DoctorID, DoctorName, Department, Address, Salary)
   Patients (PatientID, Patent Name, Address, Age, Gender)
   Hospitals (PatientID, Doctor ID, HostpitalName, Location)

   i. Display name of female Patient admitted to hospital at Pokhara.
   ii. Delete the record of Doctors whose salary is greater than average salary of doctors.
   iii. Increase the salary of doctors by 18.5% who are not from Kathmandu.
   iv. Find the average salary of Doctors for each address who have average salary more than 55K.

b) Write the SQL statements for the following queries by reference of Hotal-details relation. 7

hotel_id	hotel_name	est_year	hotel_star	hotel_worth
1	Hyatt	2047	5	1500000
2	Hotel Ktm	2043	3	500000
3	Fulbari	2058	5	2000000
4	Yak and Yeti	2052	4	1100000
5	Hotel Chitwan	2055	3	700000

i. Create a database named hotel and table relation.

ii. Create a view named Price which shows hotel name and its worth.

iii. Modify the data so that Hotel Chitwan is now four star level.

iv. Delete the records of all hotels having worth more than 900000.

3. a) Explain how you would normalize an e-commerce database that tracks customers, orders, and products. 7

b) Explain Data Constraints and its types with examples. 8

4. a) What is multi-valued dependency? Illustrate the advantage of 4NF with suitable example. 8

b) As a database administrator, discuss how Access control and Authentication can be implemented across database systems. 7

5. a) Define query optimization. What are the basic steps of query processing with the help of a diagram. 8

b) What are different recovery scheme? Explain deferred database modification with suitable examples. 7

6. a) Construct a B+-tree for the following set of key values: 7

(8, 10, 12, 14, 115, 16, 20, 24, 28, 34)

Assume that the tree is initially empty and values are added in ascending order. Construct B+ trees for the case where the number of pointers that will fit in one node is **Four.** Also show the form of the tree after deletion of **'24'**.

b) Explain the Two-Phase Locking (2PL) protocol and how it guarantees serializability in concurrent transactions. 8

7. Write short notes on: (**Any two**) 2×5

a) Dense and Sparse index

b) Trigger

c) Distributed Database

DBMS-2024 Fall

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Database Management System (New)

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between schema and instances. Explain data independence with its importance. 8

   b) Design an ER diagram for an Online Food Delivery System to manage customers, restaurants, orders, and delivery. Draw the ER diagram with cardinalities and constraints. 7

2. a) What is relational algebra? Explain Selection, Projection, Union, Cartesian-product, Join and Set-Difference operations with suitable examples. 7

   b) Consider the given relational schema: 8

   Movie(Title, Director, Myear, Rating)

   Actors(Actor, Age)

   Acts(Actor, Title)

   Directors (Director, DirAge)

   Write the SQL statement for the following task:

   i. Find the movies made by Gopal Varma after 1997.

   ii. Find all actors and directors.

   iii. Find number of movie which are directed by Devid Dhaban having actor "Bishnu".

   iv. Find (Director, Actor) pairs where the directors is younger than the actor.

   v. Find the number of movie actor wise.

3. a) Explain functional dependency with its types? For R ={A, B, C, D} and F={A->BC, B->C, AB->C, AC->D}, List the closure of functional dependency F. 7

OR

What are database constraints? Explain its types with suitable examples. When do we need Denormalization? Explain.

b) Define normalization. Create a relation in Unnormalized Form (UNF) and normalize it up to the Second Normal Form (2NF). 8

4. a) Why security is needed in database? Explain access control mechanisms. 7

OR

As a database administrator, purpose the measures to protect the database against physical and cyber threats.

b) Define ACID properties. Explain view serializability with suitable examples. 8

5. a) What do you mean by query processing? Explain the steps of query processing with necessary diagram. 7

b) Construct a B+ tree for the following set of key values: {2,5,7,9,11,17,21,23,29,31}. Assume that the tree is initially empty and values are added in ascending order where the pointer number is FOUR. 8

6. a) Consider following two transactions T1 and T2 given in figure where T1 executes before T2. Also consider initial value of A, B and C are 1000, 2000 and 3000 respectively. 7

T1	T2
Read (A)	Read (C)
A = A - 500	C = C – 200
Write (A)	Write (C)
Read (B)	
B = B + 500	
Write (B)	

Use the concept of deferred and immediate database modification log based recovery method in given problem.

b) Define data warehousing and explain how it supports decision-making processes in large organizations. 8

7. Write short notes on: **(Any two)** 2×5

a) Query cost estimation

b) Set operations in SQL

c) NoSQL Databases and its characteristics

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2021
Programme: BE Full Marks: 100
Course: Applied Operating System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define operating system. How operating system creates abstraction? Explain with reference to operating system as extended machine. 7

   b) What is process and process state? Do you think a process can exist without any state? Justify your answer with the help of a process state diagram and PCB. 8

2. a) Why semaphores are used? Give solution to reader-writer problem using semaphores. 7

   b) Suppose 5 batch jobs A, B, C, D and E arrived at the service center at time 0. They have burst time 20, 22, 8, 12 and 14 respectively. Their priorities are 4, 2, 1, 3 and 5 with 5 being the highest priority. For each of the scheduling algorithm determine average waiting time (WT) and average turn-around time (TAT) using: 8

      i. SJF
      ii. RR (Quantum size 10)
      iii. Priority (preemptive)

3. a) Consider following snapshot of a system 8

Processes	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	0	1	0	0	1	1	5	2
P1	1	0	0	1	7	5			
P2	1	3	5	2	3	5			
P3	0	6	3	0	6	5			
P4	0	0	1	0	6	5			

Answer the following questions using Banker's algorithm:

i) What is the content of Need Matrix ? 7

ii) Is the system in a safe state? Also find the safe sequence.

b) Explain different multithreading operating system designs with advantages and disadvantages.

4. a) What is mutual exclusion? Show how mutual exclusion can be achieved using Dekkers Algorithm? 7

b) How many page faults occur for following reference strings for 4 page frames? 8

5, 0, 1, 2, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 1, 2, 0, 3

Using FIFO, LRU and Optimal page replacement algorithm

5. a) Disk request come to the disk driver for cylinder 16, 18, 12, 6, 25, 38, 7 and 36 in that order. A seek take 5 micro sec per cylinder move. How much seek time is needed for 8

i. FCFS

ii. SSTF

iii. C-SCAN (upward)

iv. C-LOOK (downward)

In all case, the arm is initially at cylinder 18.

b) What is difference between Paging and Segmentation? What are different page-table structures? Explain any one in detail. 7

6. a) Describe Interrupt. How Operating system handles the interrupt? Explain with the help of a block diagram 8

b) What is file descriptor? Discuss about different file allocation methods. 7

7. Write short notes on: (**Any two**) 2×5

a) Stable storage Implementation

b) File Protection

c) RAID

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Applied Operating System

Semester: Spring

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is system call? Define major services of operating system. 8

   b) What is CPU bound and I/O bound process? Why inter process communication is required? Explain. 7

2. a) Explain the algorithm of Resource Allocation Graph. Consider a system with three processes (P0-P2) and four allocable resources (A, B, C, D). The total four resources types in the amount as E= (4, 2, 3, 1). The current allocation matrix and request matrix are as follows. 8

   Using Banker's algorithm find:

   i) What will be the context of need matrix?

   ii) Is the system in safe state? If yes, then what is the safe state sequence?

Current Allocation Matrix					Allocation Request Matrix				
Process	A	B	C	D	Process	A	B	C	D
P0	0	0	1	0	P0	0	1	2	1
P1	2	0	0	1	P1	0	0	1	0
P2	0	1	2	0	P2	1	0	2	0

   b) For what purpose semaphores are used? Give solution to producer-consumer problem using semaphores. 7

3. a) How process differs from thread? Explain and differentiate between user and kernel thread. Draw figures to illustrate. 8

   b) What mutual exclusion, race condition and critical condition? Can Peterson's algorithm is guaranteed to solve critical condition problem. Justify your answer. 7

4. a) Define page fault. How many page fault occur for the following reference string for four page frame using Optimal page replacement and LRU algorithms? 8

reference string : 1,3,1,5,7,7,3,7,4,9,8,1,6,3,4,2,5,8

b) Explain about Fixed partition and Variable partition. Consider a swapping system in which memory space is as: 7

400, 700, 1200,250,300 bytes.

File sizes are as: 900, 25, 600, 200, 300 bytes

Find the total fragmentation using

i) First Fit

ii) Next fit

iii) Best fit

iv) Worst fit

5. a) Consider the following request queue (in order): 80, 178, 24, 110, 11, 123, 61, 68 with the head initially at track 45 and the trail track being at 189. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms? 8

i. FCFS

ii. SSTF

iii. SCAN (initially moving outward)

iv. C-SCAN (initially moving inward)

b) What is fragmentation? Explain segmentation with paging with an example. 7

6. a) What are the three different ways to do input- output? Explain all. 7

b) What are different file operations? Discuss different file access methods. 8

7. Write short notes on: (**Any two**) 2×5

a) PCB

b) Interrupt Handling

c) DMA

d) File security in Linux

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2022
Programme: BE Full Marks: 100
Course: Applied Operating System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe the two basic purposes of Operating system. Differentiate time sharing system from real time system with examples. 7

   b) What is race condition and critical section? Give hardware solution to achieve mutual exclusion. 8

2. a) Producer Consumer problem can be solved with the help of various techniques. Suggest any one of the techniques to solve producer consumer with detailed explanation. 7

   b) Given the following information, draw GANTT charts and find the average waiting time and average turn-around time using FCFS, SRTN, RR (Quantum=3), HRRN. 8

Process	Arrival Time	Burst Time
A	0	9
B	3	3
C	4	4
D	6	10

3. a) What are the conditions of deadlock ? Use Banker algorithm to find whether the system is in deadlock or not? 8

Process	Allocation A B C	Maximum A B C	Available A. B. C
P1	0 1 0	7 5 3	3 2 3
P2	2 0 0	3 2 2	
P3	3 0 2	9 0 2	
P4	2 1 1	2 2 2	
P5	0 0 2	4 3 3	

   b) When multiple processes need to co-operate, there is a choice between 7

shared memory and message passing. Compare and contrast these techniques. What is the role of the operating system in each?

4. a) What is semaphore? State and give solution to dining philosopher problem. 7

   b) Given references to the following pages by a program 0,4, 1, 4, 2, 4, 3, 4, 2, 4, 0, 4, 1, 4, 2, 4, 3, 4,. How many page faults will occur if the program has four page frames for each of the following algorithms? 8

   a. FIFO. b. Optimal. c. LRU d. Second chance

5. a) Discuss in detail the use of TLB in the process of paging. Support your answer with the help of a diagram. 7

   b) Consider the following request queue (in order): 72, 171, 26, 109, 10, 125, 62, 67 with the head currently at track 50 and the previous head track was at 45. Starting from the current head position, what is the total head movement to satisfy all the pending requests, for each of the following disk-scheduling algorithms? Assume disk size 180 cylinders. 8

   i. FCFS
   ii. SSTF
   iii. SCAN
   iv. LOOK

6. a) Explain Programmed Driven I/O and Interrupt Driven I/O with Suitable diagrams. 7

   b) What are different file attributes? Explain file system allocation. 8

7. Write short notes on: (Any two) 2×5

   a) Sector slipping v and Sector Sparing.
   b) RAID
   c) Directory Structures

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Applied Operating System | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define time sharing, parallel and real-time operating systems? How they are different from one another? Explain. 8

   b) Describe the fields in a process control block (PCB) with diagram. What is switching overhead? 7

2. a) Define Semaphores. Explain solution to the Dining Philosophers Problem using semaphores. 7

   b) Given the following information's, draw GANTT charts and find the average waiting time and average turn-around time using STRF, RR (quantum=4), HRRN. 8

Process	Arrival Time	Burst Time
A	0	10
B	3	4
C	6	6
D	7	9

3. a) How many page faults occur for the following reference strings for 3-page frames using Optimal, LRU, and Second Chance.? 8
   4, 6, 7, 8, 7, 4, 8, 6, 9, 6, 4, 6, 7, 8. Also calculate fault ratio.

   b) What do you mean by disk management? Compare sector Slipping and Sector Sparing in bad block recovery during disk management. 7

4. a) What is deadlock? Consider the following snapshots of the system having 10 instances of resource A, 5 instances of resource B and 7 instances of resource C. 8

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

Answer the following questions using the Banker's Algorithm:

i. Is the system in a safe state? Also find the safe sequence.

ii. If the request from process P1 arrives for (1, 0, 2) can the request be granted immediately?

b) What is Memory Fragmentation? What are its types? How can the problem of fragmentations be solved? 7

5. a) Suppose a disk drive has 300 tracks, numbered 0 to 299. The current position of the R/W head is at track 160 and the previous request was at track 145. The sequence of pending requests is 43,172, 150, 48, 85, 270, 290, 130. Starting from current position what is the total number of track movements (distance) for the following disk scheduling algorithms. 8

i. SSTF

ii. SCAN

iii. C-LOOK

iv. FCFS

b) Explain interrupt driven I/O with diagram. How is it different from polled I/O? 7

6. a) What are different file attributes? Discuss different file allocation methods. 8

b) What is File? Compare the characteristics of various file access methods. 7

7. Write short notes on: **(Any two)** 2×5

a) RAID

b) System Calls

c) Deadlock vs. Starvation

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Applied Operating System

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time :3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are main purposes of Operating Systems? Can OS be called as a resource manager and an extended machine? Justify. 8

   b) What is a process? Explain the different states of process with a neat diagram. 7

2. a) Define semaphores. How Dining Philosopher problem can be solved using semaphores? Explain with necessary pseudo code. 7

   b) Consider following set of processes along with their burst time, arrival time and priorities. Calculate average waiting time and average turnaround time using following scheduling. Also describe which one is the best algorithm and why? 8

      i. SRTF
      ii. Priority
      iii. RR (Quantum size = 3ms)

Process	Burst Time	Arrival Time	Priority
P1	3	0	2
P2	14	1	1
P3	9	2	3
P4	17	3	4

3. a) Consider following snapshot of a system 8

Processes	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
A	0	1	1	1	2	3	0	1	0
B	1	1	0	2	2	0			
C	0	0	1	0	1	1			
D	1	2	1	3	5	3			
E	1	0	1	1	1	2			

Answer the following questions using Banker's algorithm:

   i. What is the content of the Need Matrix?
   ii. Is the system in a safe state? Also find the safe sequence.

b) Define thread. Why is it called a light weight process? Compare and contrast between process and thread. 7

4. a) Explain race condition and critical condition. Can Peterson's algorithm be guaranteed to solve critical condition problem. Justify your answer. 7

b) What is page fault? Given below is the reference made to the following pages by a program: 1,3,1,5,7,7,3,7,4,9,8,1,6,3,4,2,5,8. Show the successive page residing in the four frames using replacement policy below: 8

i. Second Chance
ii. LRU
iii. Optimal

5. a) The disk track requests are: 111, 123, 250, 298, 120, 13, 288 and 224. There are 300 cylinders numbered from 0 to 299. Assume that the last request is at track 130 and the head is moving towards track 0. Find out the total seek time for each of the disk scheduling algorithms below: 8

i. SSTF
ii. C-SCAN
iii. LOOK

b) Difference between Fixed partition and Variable partition. Consider a swapping system in which memory space is as: 400, 750, 1200,250,150 bytes. 7

File size are as: 200, 75, 600, 900,300

Find the total fragmentation using

i. First Fit
ii. Next fit
iii. Best fit
iv. Worst fit

6. a) What are the problems of programmed and interrupt driven I/O techniques? How does DMA solve these problems? Explain in detail. 8

b) Explain all methods used for file allocation. Compare and contrast all the methods with appropriate diagrams. 7

7. Write short notes on: **(Any two)** 2×5

a) Free space management
b) Scan vs CScan
c) Stable Storage Implementation

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Applied Operating System

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define OS as a resource manager and an extended machine. Differentiate between batch system and time-sharing system. 8

   b) Explain the lifecycle of a process with the help of a process state diagram. 7

2. a) Define Semaphores. Explain solution to the Producer Consumer Problem using semaphores. 7

   b) Given the following information's, draw GANTT charts and find the average waiting time and average turn-around time using FCFS, RR (quantum=3), SRTF, HRRN. 8

Process	Arrival Time	Burst Time
P1	0	9
P2	3	4
P3	4	5
P4	6	10

3. a) What is deadlock? Find the safe sequence if any, using the Bankers Algorithm for the following information. 8

Process	Allocation	Max	Available
	A B C D	A B C D	A B C D
P0	6 0 1 2	4 0 0 1	3 2 1 1
P1	2 7 5 0	1 1 0 0	
P2	2 3 5 6	1 2 5 4	
P3	1 6 5 3	0 6 3 3	
P4	1 6 5 6	0 2 1 2	

   b) Explain race condition and critical condition. Can Peterson's algorithm is guaranteed to solve critical condition problem. Justify your answer. 7

4. a) Difference between Fixed partition and Variable partition. Given the following free memory blocks: 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB. If a new process requires 212 KB, 350 KB, 525 KB, 85 KB use the first-fit, best-fit, next fit and worst-fit memory allocation strategies to determine which block will be allocated to the process. Also, calculate the resulting total fragmentation for each strategy. 7

b) Define page fault. How many page faults occur for the following reference strings for 3-page frames? 5, 6, 7, 8, 7, 5, 8, 6, 9, 6, 5, 6, 7, 8. Using Optimal, Second Chance, and LRU. Also calculate fault ratio. 8

5. a) Consider a disk with 200 tracks (numbered from 0 to 199). The disk head is initially at track 50. The requested tracks, in the order they arrive, are: 95, 180, 34, 119, 11, 123, 62. Calculate the total head movement using: 8

   i. C-SCAN
   ii. FCFS
   iii. SSTF
   iv. LOOk

b) What are different ways to input output? Explain DMA with suitable diagram. 7

6. a) What is File? Explain i-node file system implementation. 7

b) What are different file attributes? Explain Contiguous and Linked List methods of file allocation. 8

7. Write short notes on: **(Any two)** 2×5

a) Swap- space Management

b) Context Switching

c) RAID

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Applied Operating System (New)

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain Operating System. List out different types of operating system. Explain any three of them in brief. 8

   b) What is a process? Illustrate and define the different states of process with a neat diagram. 7

2. a) Define Critical Section and Race Condition. How Dinning Philosopher problem can be solved using semaphore? 7

OR

What are the usage of sleep and wakeup operations in process synchronization? Explain two-task solution to achieve mutual exclusion.

   b) Explain Resource Allocation Graph. Consider a system with three processes P1 through P3 and four allocable resources. The total four resources types in the amount as E=(4,2,3,1). Suppose at time t0 following snapshot of the system has been taken: 8

   i. What is the content of the need matrix?
   ii. Is the system in safe state?

Process	Allocation Matrix				Process	Max Matrix			
	R1	R2	R3	R4		R1	R2	R3	R4
P1	0	0	1	0	P1	2	0	1	1
P2	2	0	0	1	P2	1	0	0	0
P3	0	1	2	0	P3	2	1	0	0

3. a) What is page fault? Given below is the reference made to the following pages by a program: 1,3,1,5,7,7,3,7,4,9,8,1,6,3,4,2,5,8. Show the successive page residing in the four frames using replacement policy below: 8

   i. Second Chance
   ii. LRU
   iii. Optimal

b) Define file system. Explain linked list and i-node file system implementation. 7

4. a) Differentiate sequential file access method and random file access method. Explain different file allocation strategies. 7

b) Explain Direct Memory Access mechanisms. Explain the situations where DMA and programmed IO performed better than each other. 8

OR

Define device drivers. Explain three different ways of performing input and output.

5. a) The disk track requests are: 125, 250, 298, 123, 13, 350 and 224. Assume that the last request is at track 150 and the head is moving towards track 0. Find out the total seek time for each of the disk scheduling algorithms below: 8

i. SSF
ii. C-SCAN
iii. FIFO
iv. LOOK

b) What are the different components used in Windows Operating System. 7

6. a) Given the following set of information, calculate the average waiting time and average turn-around time using SJF(preemptive), RR (Quantum = 2) and priority(smaller number as higher priority) scheduling. Also describe which one is the best algorithm and why? 8

Process	Burst Time	Arrival Time	Priority
P1	8	0	3
P2	1	1	1
P3	3	2	2
P4	2	3	3
P5	6	4	4

b) What are the characteristic features of distributed operating system? Explain cloud operating system. 7

7. Write short notes on: **(Any two)** 2×5

a) Virtual Machine
b) Swapping
c) RAID

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Applied Operating System

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is OS and what are its main responsibilities? Explain OS as an user extended machine with suitable diagram. 8

   b) Explain the lifecycle of a process with the help of a process state diagram. 7

2. a) Consider following set of processes along with their burst time, arrival time and priorities. Calculate average waiting time and average turnaround time using following scheduling. Also describe which one is the best algorithm and why? 8

   i. SJF (Preemptive)
   ii. Priority
   iii. RR (Quantum size = 3ms)

Process	Burst Time	Arrival Time	Priority
P1	7	0	2
P2	7	0	1
P3	6	2	3
P4	5	5	4

   b) Define Semaphores. Explain producer consumer problem and solution using semaphores. 7

3. a) What is deadlock? Find the safe sequence if any, using the Bankers Algorithm for the following information. 8

Process	Allocation	Max	Available
	A B C D	A B C D	A B C D
P0	6 0 1 2	4 0 0 1	3 2 1 1
P1	2 7 5 0	1 1 0 0	
P2	2 3 5 6	1 2 5 4	
P3	1 6 5 3	0 6 3 3	
P4	1 6 5 6	0 2 1 2	

b) Explain page fault. Given below is the reference made to the following pages by a program: 2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5, 2, 1, 4, 6, 2, 3. Show the successive page residing in the three frames using replacement policy below: 7

i. FIFO

ii. LRU

iii. Optimal

4. a) Differentiate between internal and external fragmentation? Explain TLB in detail. 7

b) Difference between Fixed partition and Variable partition. Given the following free memory blocks: 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB. If a new process requires 212 KB, 350 KB, 525 KB, 85 KB use the first-fit, best-fit, next fit and worst-fit memory allocation strategies to determine which block will be allocated to the process. Also, calculate the resulting total fragmentation for each strategy. 8

5. a) Explain interrupt driven I/O with diagram. How is it different from polled I/O? 7

b) Consider the following request queue (in order): 72, 178, 34, 109, 10, 135, 58, 89 with the head currently at track 34. Starting from the current head position, what is the total head movement to satisfy all the pending requests, for each of the following disk-scheduling algorithms? Assume disk size 180 cylinders. 8

i. C-SCAN

ii. SSTF

iii. LOOK

iv. FCFS

6. a) Explain Kernel I/O subsystem. 8

b) What are different file attributes? Discuss about different free space management strategy in file systems. 7

7. Write short notes on: **(Any two)** 2×5

a) System Calls

b) File Access Methods

c) Resource Allocation Graph

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Applied Operating Systems (New)

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain Operating System. List out different types of operating system. Explain any three of them in brief. 7

   b) Why process is heavyweight in comparison with thread? How Process Control Block (PCB) is used in process management and to handle context switching between processes? 8

2. a) Draw a Gantt Chart and find average turnaround time and waiting time of the following process applying SJF(Preemptive), SRTF, FCFS and Round Robin (quantum=3) scheduling algorithm. 8

Process	A	B	C	D	E
Arrival Time	0	4	4	3	2
Burst Time	4	6	5	8	7

   b) Explain the concept of semaphores as synchronization primitives. Explain how mutual exclusion and synchronization ensure correct execution in concurrent environments. 7

OR

What is race conditions and how they can lead to inconsistent or incorrect results in shared data access? Explain the basic requirement for the solution of critical section problem.

3. a) Define the critical section. The snapshot of the state is given below. Is this a safe state? If yes, what is its safe sequence? 8

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	3	0	1	4	5	1	1	7	1	0	0	2
P1	2	2	1	0	3	2	1	1				
P2	3	1	2	1	3	3	2	1				
P3	0	5	1	0	4	6	1	2				
P4	4	2	1	2	6	3	2	5				

b) Differentiate sequential file access method and random file access method. Explain different file allocation strategies. 7

4. a) What is page fault? Given below is the reference made to the following pages by a program: 1,3,1,5,7,7,3,7,4,9,8,1,6,3,4,2,5,8. Show the successive page residing in the four frames using replacement policy below: 8

i. FIFO

ii. LRU

iii. Optimal

Also calculate page fault.

b) What is virtual memory? Explain the structure of page table with suitable diagram. 7

OR

Under what circumstances do page faults occur? Describe the actions taken by the operating system when a page fault occurs.

5. a) The disk track requests are: 125, 250, 298, 123, 13, 350 and 224. Assume that the last request is at track 150 and the head is moving towards track 0. Find out the total seek time for each of the disk scheduling algorithms below: 8

i. SSF

ii. C-SCAN

iii. FIFO

iv. SCAN

b) Define device controllers. Explain I/O software layers. 7

6. a) Discuss the design principles IPC mechanisms and security management of any operating system (LINUX, WINDOWS). 7

b) Explain the concept of the memory wall and discuss the major bottlenecks faced by operating systems in memory management. 8

7. Write short notes on: **(Any two)** 2×5

a) OS as a virtual machine

b) Goals of protection

c) LINUX Vs Windows OS

AOS (New) 2024Fall

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Microprocessor and Computer Architecture | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) How is a computer organized? Explain Von Neumann Architecture and Harvard Architecture with necessary diagrams. 7

   b) Draw the block diagram of an 8085 microprocessor and explain, in brief, all the functional units in it. 8

2. a) How the instructions are classified in 8085 microprocessor? Explain with examples 8

   b) What is the need for timing diagram Draw the timing diagram of instruction ***MOV A,B*** 7

3. a) Write the features of 8086 microprocessor. Explain the addressing modes in 8086 Microprocessor. 7

   b) Design a very simple CPU, with its state diagram, RTL codes, register section and ALU, for the given set of instructions. The system uses 6-bit Address register, 6-bit Program counter, 8-bit Data register and 2-bitInstruction register. 8

Instruction	Instruction Code	Operation
ADD	00AAAAAA	AC←AC + M[AAAAAA]
OR	01AAAAAA	AC←AC v M[AAAAAA]
JMP	10AAAAAA	GOTO AAAAAA
INC	11XXXXXX	AC←AC+2

4. a) Design hardwired control unit for the CPU described in question 3b. Also, briefly explain its operation, how the control signals are generated and execution performed. 7

   b) Explain the design of a micro-sequencer control unit with vertical microcode for the CPU described in question 3b. 8

5. a) Define RTL. Mention the RTL codes for shift-add algorithm and trace the multiplication of 13 and 11. Also, verify the obtained product. 8

OR

How signed numbers are represented? Write RTL codes for Booth's algorithm and use it to multiply -13 and 3.Also, verify the obtained product.

b) Differentiate between Segmentation and Paging? Describe the two most common replacement algorithms related to design issues in Cache memory. 7

6. a) What do you understand by DMA? Explain it with suitable diagram. 8

OR

Describe with the proper diagram the working principle of I/O processors.

b) What are the different topologies used to interconnect MIMD computers? Describe with suitable diagrams. 7

7. Write short notes on: (Any two) 2×5

a) A 3×3 Wallace Tree

b) Multicore processor architecture

c) Microprocessor Vs Micro controller

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessor and Computer Architecture

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Difference between Von-Newmann and Harvard architecture with necessary diagram. 7

   b) What are the different types of FLAGS used in 8085 microprocessor? Show the status for the Flags after performing the following operation 5F H+ 02 H. 8

2. a) Explain the timing diagram for ORI 07H. 8

   b) Write an ALP for the following addition. 7

   $1^2+2^2+3^2+4^2+5^2+6^2+7^2+8^2+9^2$

3. a) What are the addressing modes of 8086 microprocessor? Explain in detail. 7

OR

Explain ALP development tools in 8086 with examples.

   b) Design a very simple CPU with given instruction set: 8

Instruction	Instruction Code	Micro-operations
ADD	00 AAAAAA	AC← AC+M[AAAAAA]
JMP	01 AAAAAA	PC← AAAAAA
OR	10 AAAAAA	AC← AC$^{\vee}$M[AAAAAA]
SKIP	11 XXXXXX	PC← PC+1

Write down the RTL codes for the CPU, draw state diagram, and register section and ALU.

4. a) Design a hardwired control unit for the CPU given in question number 3b. 7

   b) Design a micro-sequencer control unit with horizontal microcode for the given CPU in question number 3. b. 8

5. a) Draw the flowchart for the Booth's algorithm and perform -9×-9. 7

OR

Show the division of the number 14/2 using Restoring Division algorithm.

b) What is memory hierarchy? Differentiate between direct, associative and set associative mapping. 8

6. a) Draw and explain the flow of Interrupt I/O during input/output transfer. How is it more efficient than programmed I/O? 7

b) What are the major conflicts occurring due to instruction pipelining in RISC? Explain. Describe the solutions to correct data conflicts. 8

7. Write short notes on: **(Any two)** 2×5

a) Flynn's taxonomy

b) HDMI

c) Instruction set

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2024
Programme: BE | Full Marks: 100
Course: Microprocessor and Computer Architecture | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between Microprocessor and Microcontroller with examples. 7

   b) What are the different types of FLAGS used in 8085 microprocessor? Show the status for the Flags after performing the following operation 5F H+ 02 H. 8

2. a) What is machine cycle? Explain the timing diagram of an instruction LXI H, 3050H. 7

   b) Write a 8085 ALP to find the largest number from an array of given 10 numbers stored at location starting from COOOH and store the result in the memory location 2024H. 8

OR

Write a 8085 ALP to find smallest of three number.

3. a) Write an ALP to find the smallest number in a data array. Data from location C000H to C005H. 7

OR

Explain ALP development tools in 8086 microprocessor.

   b) What There is a very simple CPU for the given set of Instructions: 8

Instructions	Instruction Code	Operations
OR	00AAAAAA	AC←AC∨M[AAAAAA]
AND	01AAAAAA	AC←AC ∧ M[AAAAAA]
CMP	10AAAAAA	AC←AC'
LDA	11XXXXXX	AC← M[AAAAAA]

Let the instruction width be 8-bits and address is 6 bits. Design the CPU's Register Section, State Diagram and ALU.

4. a) Design the hardwired control unit for the CPU described in question 3 b. 7

	b)	Design a micro-sequencer control unit with horizontal microcode for the given CPU in question number 3b.	8
5.	a)	Trace the multiplication of 4 and -7 using RTL code of Booth's algorithm. Also verify the obtained product.	8
	b)	Explain the Set- Associative technique during the mapping of Cache memory and compare it with associative mapping.	7
6.	a)	How DMA reduces the CPU engagement dealing I/O interrupts with DMA than programmed I/O and interrupt driven I/O. Explain.	7
	b)	Describe Flynn's Taxonomy. Explain the different topologies used in multiprocessor communication.	8
7.	Write short notes on: (Any two)		2×5
	a)	RISC Vs. CISC	
	b)	Hardwired Vs. micro programmed CU	
	c)	Computer Organization Vs. Computer Architecture	

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Applied Mathematics

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define analyticity of a function f(z). Show that the necessary condition for the function $f(z) = u(x,y) + iv(x,y)$ to be analytic on a domain D is $u_x = v_y$ and $u_y = -v_x$ at each point (x,y) of D. 7

   b) State and prove Cauchy integral formula. Evaluate the integral $\oint_c \frac{e^{5z}}{(z+i)^4} dz$, where c: $|z| = 2$. 8

OR

State Cauchy's residue theorem and using it, evaluate:
$\oint_c \frac{2z}{(z+1)(z-1)^3(z+3)}\, dz$ where c:$|z| = 2$ counter clockwise.

2. a) Find the expansion of $\frac{7z-2}{z(z+1)(z-2)}$ in the region given by 7

   i. $0 < |z+1| < 1$.

   ii. $1 < |z+1| < 3$.

   b) Define bilinear transformation. Find the bilinear transformation which maps the points $z = 0, -1, i$ onto the points $w = i, 0, \infty$. Also, find the image of the unit circle $|z| = 1$. 8

3. a) State and prove first shifting theorem on Z-transform. Find Z-transform of $e^{\frac{in\pi}{2}}$ and then find $Z(\cos\frac{n\pi}{2})$ and $Z(\sin\frac{n\pi}{2})$. 7

   b) Solve the difference equation $y_{n+2} - 7y_{n+1} + 12y_n = 2n$, $y_0 = 0$, $y_1 = 0$ by using Z-transform. 8

4. a) Show that $Z[nf(t)] = -z\frac{d}{dz}[F(z),]$ where $F(z) = Z[f(t)]$. 7

   Find $Z^{-1}\left[\frac{z}{(z+1)^2(z-1)}\right]$.

   b) Find the solution of one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}$ with initial velocity $g(x)$, initial deflection $f(x)$ and boundary condition u(0, L) = 0 = u(L, t). 8

5. a) Express the Laplacian $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ in polar coordinates. 7

OR

Derive two-dimensional heat equation completely with necessary assumptions.

b) Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature 0, assuming that the initial temperature is $f(x) = \begin{cases} x & \text{if } 0 < x < \frac{L}{2} \\ L - x & \text{if } \frac{L}{2} < x < L \end{cases}$. 8

6. a) Find the Fourier cosine transform of $f(x) = e^{-mx}$ for $m > 0$, and then show that $\int_0^\infty \frac{\cos kx}{1+x^2} dx = \frac{\pi}{2} e^{-k}$. 7

b) Show that $\int_0^\infty \frac{\cos wx + w \sin wx}{1+w^2} dw = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$. 8

7. Attempt all the questions: 4×2.5

a) Check analyticity of $f(z) = z^2$.

b) Find $Z(a^n)$.

c) Find the solution of the partial differential equation $u_{xx} + 9u = 0$.

d) Define linear partial differential equation with suitable example.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Applied Mathematics

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define analyticity of a function f(z). Show that the necessary condition for the function $f(z) = u(x,y) + iv(x,y)$ to be analytic on a domain D is $u_x = v_y$ and $u_y = -v_x$ at each point (x,y) of D. 7

   b) State and prove Cauchy integral formula. Evaluate the integral $\oint_C \frac{e^{5z}}{(z+i)^4} dz$, where c: $|z| = 2$. 8

2. a) State Cauchy Residue Theorem. By applying Cauchy Residue Theorem, evaluate $\oint_C \frac{dz}{(z^2+4)^3}$ where $C : |z - i| = 2$. 8

   b) Find the linear transformation that maps $z = 0$, $z = 1$, $z = \infty$ into the points $w = -3$, $w = -1$, $w = 1$ and find the fixed point of the transformation. 7

3. a) State and prove first shifting theorem on Z-transform. Find Z-transform of $e^{\frac{in\pi}{2}}$ and then find $Z(\cos\frac{n\pi}{2})$ and $Z(\sin\frac{n\pi}{2})$. 7

   b) Solve the difference equation $y_{n+2} - 7y_{n+1} + 12y_n = 2n$, $y_0 = 0$, $y_1 = 0$ by using Z-transform. 8

4. a) Show that $Z[nf(t)] = -z\frac{d}{dz}[F(z),]$ where $F(z) = Z[f(t)]$. 7

   Find $Z^{-1}\left[\frac{z}{(z+1)^2(z-1)}\right]$.

   b) Derive one dimensional heat equation. 8

5. a) Change the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ into polar form $r^2 u_{rr} + r\, u_r + u_{\theta\theta} = 0$. 8

   b) The diameter of a semi-circular plate of radius 'a' is kept at 0°C and the temperature at the semi-circular boundary is T°C. Find the steady state temperature of the plate. 7

OR

Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature 0, assuming that the initial temperature is $f(x) = \begin{cases} x & \text{if } 0 < x < \frac{L}{2} \\ L - x & \text{if } \frac{L}{2} < x < L \end{cases}$.

6. a) Find the Fourier cosine transform of $f(x) = e^{-mx}$ for $m > 0$, and then show that $\int_0^\infty \frac{\cos kx}{1+x^2} dx = \frac{\pi}{2} e^{-k}$. 7

OR

Find the Fourier transform of $f(x) = xe^{-x^2}$.

b) Show that $\int_0^\infty \frac{\cos wx + w \sin wx}{1+w^2} dw = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$. 8

7. Attempt **any two** questions. 2×5

a) Find Laurent series of the function $f(z) = \frac{z+3}{z(z+1)(z-2)}$ in the region $1 < |z| < 2$.

b) Find the solution of the differential equation, $y^2 u_x - x^2 u_y = 0$, by using separating of variables.

c) Show that the function $u(x, y) = e^{2x}(x \cos 2y - y \sin 2y)$ is a harmonic function. Find its harmonic conjugate.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: System Administration in IT Infrastructure Services

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define System Administration. Discuss the importance of System Administration in IT Infrastructure. 7

   b) What do you mean by Service Level Agreements and its types. Explain IT service management Frameworks. 8

2. a) Discuss the structure of the Linux file system. Explain the concepts of user and group management in Linux. 8

   b) Provide an overview of Windows Server Operating Systems and explain the role of Active Directory and Domain Controllers. 7

3. a) What is subnetting, and why is it important in network management? Given the IP address 192.168.100.0 with a subnet mask of 255.255.255.0 (/24), create 4 subnets. Calculate the range of each subnet and determine the Network address and Broadcast address for each subnet. 7

   b) Describe the four layers of the TCP/IP model and their respective functions. How does the TCP/IP model differ from the OSI model, and what similarities do they share? Explain in brief. 8

   OR

   Explain the principles of information security and identify common security threats and vulnerabilities.

4. a) List out the differences between virtualization and physical infrastructure. Explain in brief about different types of virtualizations. 7

   b) Explain the different Cloud computing models and services in brief. 8

5. a) Describe the challenges of virtualization in modern IT environments. How does virtualization contribute to cost savings and improved resource allocation? Explain in brief. 8

   b) Define DevOps and explain its core principles and practices. 7

6. a) How can you automate routine tasks using any of the scripting language? Explain with an example. 8

   b) How can organizations design and apply a structured troubleshooting workflow to improve system administration practices? Explain. 7

7. Write short notes on: **(Any two)** 2×5

   a) IT Service Management with ITIL

   b) Domain Name System (DNS)

   c) Ubuntu

POKHARA UNIVERSITY

Level: Bachelor	Semester: Fall	Year : 2024
Programme: BE		Full Marks : 100
Course: System Administration and IT Infrastructure Services		Pass Marks : 45
		Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are the key roles and responsibilities of a system administrator? Explain in detail. 7

   b) Explain how the Service Level Agreements (SLAs) and IT Service Management (ITSM) frameworks contribute to delivering high-quality IT services. 8

2. a) Explain the different types of the Linux file system. 8

   OR

   Illustrate the concepts of user and group management in Linux and the importance of file permissions and security.

   b) Explain the role of Active Directory and Domain Controllers. Also, discuss the various Windows Server roles and features. 7

3. a) Define Protocol. Discuss the function of OSI and compare with TCP/IP protocol stack. 7

   OR

   Explain the principles of information security and highlight common security threats and vulnerabilities.

   b) Describe the best practices for protecting services and networks in detail. 8

4. a) Explain the fundamental concepts and functions of file server in detail. 7

   b) Compare virtualization with physical infrastructure. Explain any two popular virtualization platforms. 8

5. a) What is cloud computing? Discuss the different SPI models with appropriate examples. 8

   b) Explain the role of scripting languages such as Bash and PowerShell in automating routine tasks. 7

6. a) Explain how DevSecOps principles and practices enhance the secured software development and deployment lifecycle. 7

   b) How can organizations design and apply a structured troubleshooting workflow to improve system administration practices? Explain in detail. 8

7. Write short notes on: **(Any two)** 2×5

   a) Qualities of Enterprise System Admins

   b) Linux Distribution

   c) Type 1 and 2 Hypervisors

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